non æquivaleant potentiæ 2BR. Duplicatis vero angulis inter BR & quamcunque BS seu BT, propositio quoque manifesta fit de illis potentiis, quæ transgrediuntur terminos BD, BE. Sicque tota demonstratio absoluta est.

Schol. Demonstrata compositione virium pro omnibus rhombis, valebit eadem pro omnibus rectangulis (per prop. 6.) atque pro omnibus parallelogrammis, per coroll. 2. prop. 3.

Geminus Modus directus *DIVIDENDI SEMICIRCVLVM* in data ratione :

Quibus Keplerianum Problema *DE INVENIENDIS PLANETARVM LOCIS* ad datum quodvis tempus,

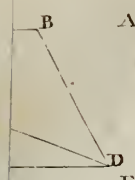
Solutum exhibetur.

Auctore

Iac. Hermanno.

25. *Febr.* 1726.

Iohannes Keplerus primus quod sciam docuit planetarum semitam ellipticam esse, & hos planetas ipsos in sua quemque orbita ita circumferri, ut areas verrant tempori motionis proportionales. Pulchra hæc reperta ex Tychonicis observationibus eruta calculo



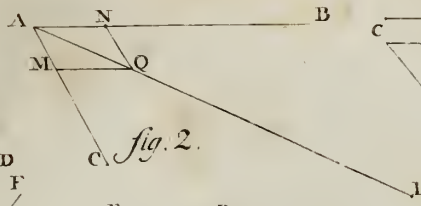


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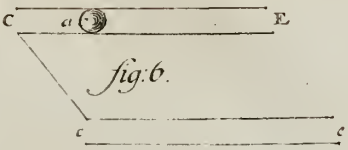


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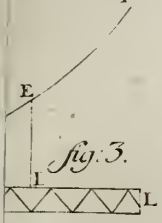


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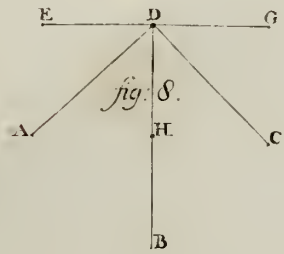


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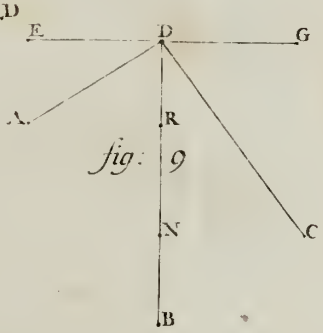


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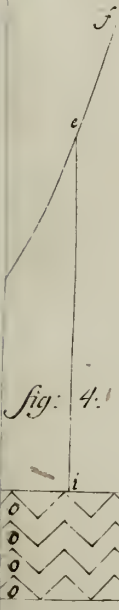


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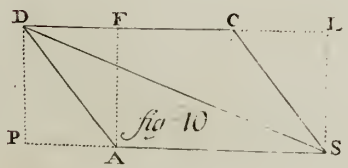


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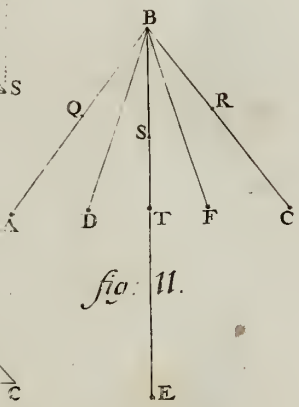


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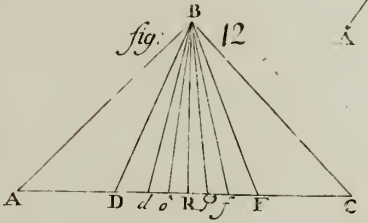


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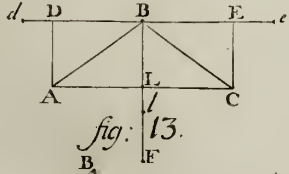


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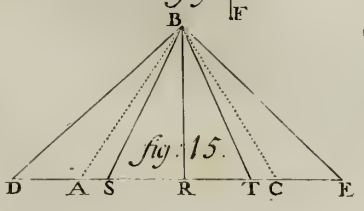


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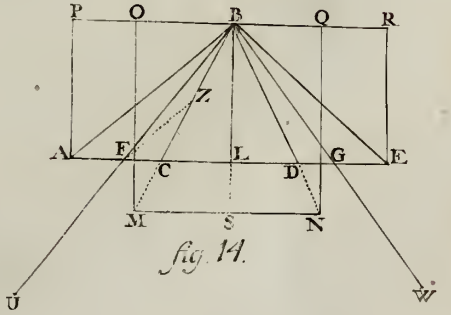
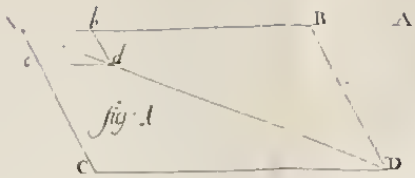
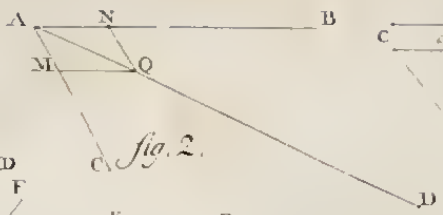
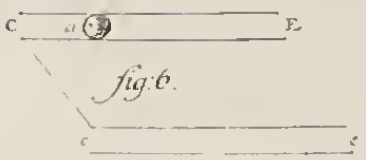
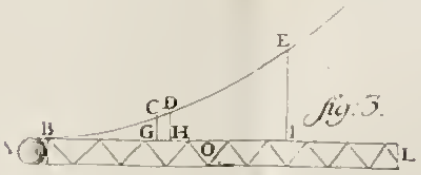


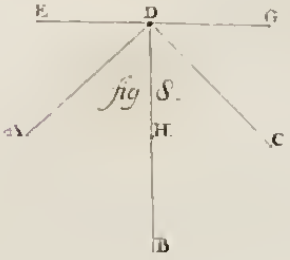
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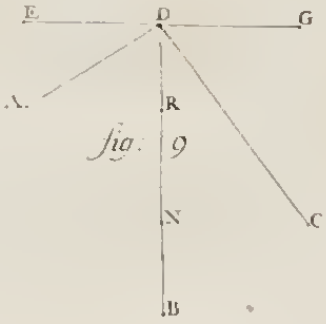

fig. 1

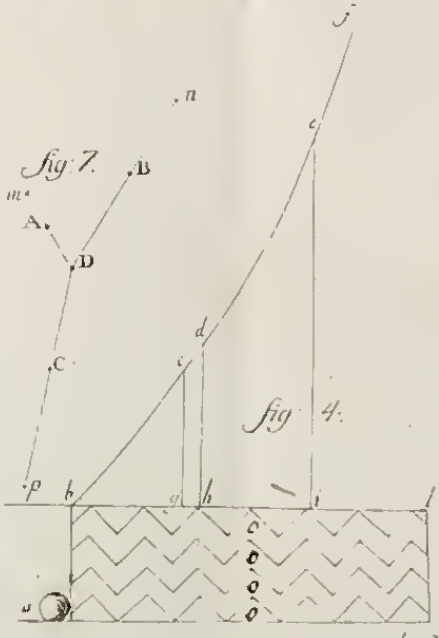

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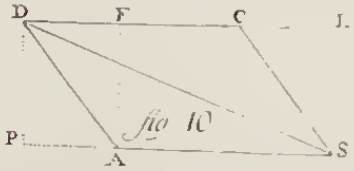

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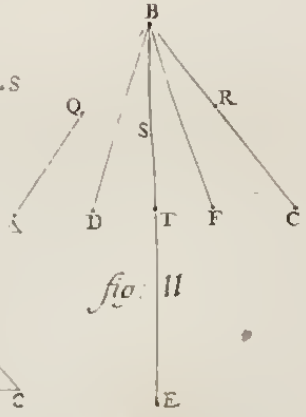

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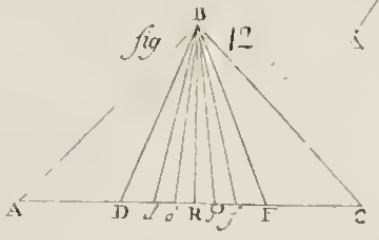

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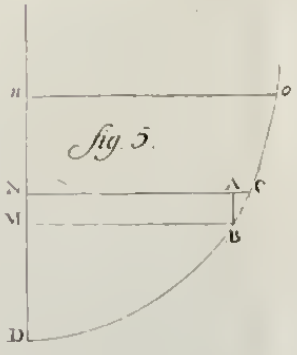

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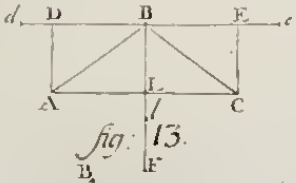

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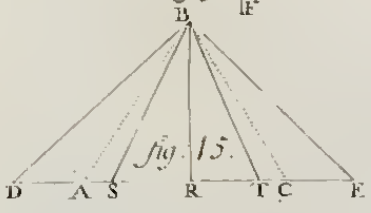

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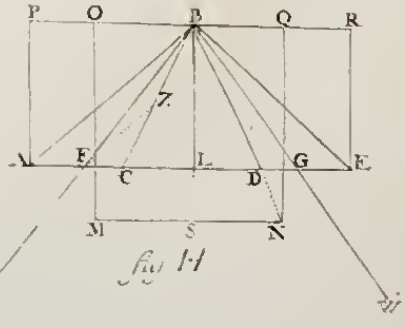

fig. 11


fig. 12


fig. 5.


fig. 13.


fig. 15.


fig. 14

lo ſagaci induſtria poſito & ſubducto, recentioribusque obſervationibus apprime conſentanea, Aſtronomiæ plane novam induxerunt faciem. In hac vero difficile ſeſe offerebat problema ſolvendum, ad cuius ſolutionem nulla via Keplero patebat directa, quamque proinde per regulam quam vocant *falſi* tentare coactus erat. Keplerus ad finem Partis IV. Commentarii in Stellam Martis fol. 300, problema his verbis enuntiat. *Data area partis Semicirculi, datoque puncto diametri invenire arcum, & angulum ad illud punctum: cuius anguli cruribus, & quo arcu, data area comprehenditur.* Vel, *Aream ſemicirculi ex quocunque puncto diametri in data ratione ſecare*: Subiungens: mihi ſufficit credere, ſolvi a priori non poſſe propter arcus & ſinus στερογενειαν. Erranti mihi, quicunque viam monſtraverit, is erit mihi magnus Apollonius.

2. Primus, quantum mihi conſtat, qui magni Apollonii laudem mereri conatus eſt, fuit *Chriſtophorus Wrennus*, qui reapſe viam directam inivit & problema ope Cycloidis protractæ conſtruxit; *Walliſius* enim conſtructionem eius ad calcem tractatus de Cycloide prodidit, an vero eam potiſſimum viam, qualem *Keplerus* ſibi monſtrari optavit, preſſerit, de eo valde dubito. Ipſe enim Cel. *Newtonus* qui eandem fere conſtructionem Prob. 31. Lib. 1. *Princip. Ph. Nat.* repetiit, faſſus eſt, propter difficilem curvæ huius deſcriptionem, præſtare ſolutionem adhibere vero proximam, qualem per ſeriem quandam angulorum convergentem exhibuit, neſcio tamen, an ſatis expeditam. Illuſtrem Geometram, ut in aliis, ita & hic imitati ſunt *David Gregorius* & *Ioh Keill* uterque enim problema Keplerianum per ſeriem infinitam ſolutum dedit, hic in Lectionibus Aſtronomicis, ille vero

vero in Elementis Aſtronomiæ Geometricis. Citra ſcrierum uſum vero eidem problemati ſolvendo incubuerunt Duumviri de Aſtronomia præclare meriti *Phil. de la Hire*, atque *Iacobus Caſſinus*, & ſolutiones ſuas in Commentariis Academiæ Reg. 1710 & 1719 ediderunt. Alia comparuit in Tomo VI. Supplementorum ad acta Eruditorum Lipſienſia, Autore *Petro Horrebow*.

3. Supervacaneum forte videbitur, quod poſteaquam hoc problema tot inſignibus autoribus ſolutum eſt, id denuo in ſcenam producere audeam. Audebo tamen, nec hoc faciens actum agere exiſtimo, cum diverſi unum idemq; problema ſolvendi modi utilitate ſua non deſtituantur. Geminum habeo, alterummere geometricum, alterum vero arithmeticum calculoque aſtronomico magis accomodum. Problema quod præ manibus habemus eſt eiusmodi.

Dividere Semicirculum AQD *per lineam* BE *ita, ut trilineum* ABE *ſit ad ſemicirculum* AQD, *in data ratione* DN *ad* DH=AD.

Conſtructio. Semicirculo DGH alteri AQD æquali inſcribatur quadratrix DLGH talis indolis, ut ex quolibet eius puncto L ductis LK parallela diametro DH, & LM eidem perpendiculari, ſit ubique arcus reſpondens huius diviſionis, nempe DK, ad abſciſſam DM, ut DKG ad radium DI. Sumatur dein in radio CQ ſurſum producto & diametro perpendiculari, CP=quadranti AQ, & iungatur PE, ex puncto vero N agatur NL alteri EP parallela, quadratrici alicubi occurrens in L, ex quo puncto deinceps ducta LB diametro AD parallela, occurret ſemicirculo AQD in optato puncto B, iungendo enim BE, triangulum ABE erit ad ſemicirculum AQD in data ratione DN ad DH. De-

Demonſtratio. Nam ex natura lineæ DLG fit AB (DK) : AQ (DG) :: DM : AC (DI), adeoque AQ×DM =AB×AC= 2 Sect. ABC. Propter parallelas PE & LN, triangula PCE & LMN ſimilia ſunt,& præbent CP (conſtr. AQ) : CE :: LM (BF) : MN, quare AQ×MN =CE×BF=2△CEB, & addendo AQ×DM+AQ× MN= 2 Sect. ABC+2△CBE=2 trilin. ABE, quare AQ×DN= 2trilin. ABE;& 2 Semic. AQDA=AQ ×AD vel =AQ×DH ; quare Trilineum ABE : Semicirc. AQDA (::AQ×DN : AQ×DH) :: DN : DH. Quod erat &c.

Curva DLG eſt quidem transcendens, cum ſit ſpecies lineæ *Sinuum*, eo tamen rectius ad conſtructionem adhibetur, quanto facilius Cycloide protracta per puncta deſcribi poteſt. Per continuam enim biſectionem quadrantis DKG eiusque partium, tot curvæ puncta & tam propinqua inveniri poſſunt,quot quis voluerit, ducendo tantum per ſingulas diviſiones quadrantis lineas diametro parallelas & per ſimiles diviſiones radii DI lineas eidem diametro perpendiculares, occurſus harum perpendicularium diametro, & illarum parallelarum, ſuppeditant totidem curvæ puncta.

Dixi curvam DLG ſpeciem eſſe lineæ *Sinuum*: nam LM eſt ſinus arcus DK & DM huic arcui proportionalis, quare abſciſſæ DM arcus,& ordinatæ LM ſinus eorum repræſentant, & hanc ob rationem curva DLG eſt *linea ſinuum*.

Hæc curva, quæ aliis Quadratrix Tschirnhauſiana vocatur, quod mentio eius occurrat in *Tschirnhauſii* Medicina Mentis, ubi aliquot eiusdem proprietates recenſentur ; hæc linea, inquam, egregium uſum habet in con-

ſtructionibus illorum problematum, quæ ab arcubus circularibus pendent, quod hæc ipſa conſtructio problematis Kepleriani probat.

Data conſtructio quoque arithmeticam problematis ſolutionem ſubminiſtrat. Nam vocando arcum AB$=a$, radium AC$=r$, ſinum arcus BF$=s$, quadrantem ABQ$=q$, & tandem DN$=b$, ac CE$=e$, propter curvam DLG fiet DM$=\frac{ar}{q}$, triangula vero ſimilia CPE, & MLN, præbent MN$=\frac{es}{q}$; quare cum ſit DM$+$MN$=b$, fiet $\frac{ar+es}{q}=b$; vel $ar+es=bq$.

Iam vero per ea, quæ *Ill. Leibnitius* in Actis Erudit. 1693 pag. 179 ex calculo differentiali, & ego occaſione problematis angularis per communem geometriam, in Actis 1703 pag. 351 elicuimus, eſt $s=a-\frac{a^3}{6rr}+\frac{a^5}{120r4}-\frac{a^7}{5040r6}+$&c. quare ſurrogando hanc ſinus æſtimationem in ſupra inventa æquatione $ar+es=bq$, habebimus $(r+e)a-\frac{a^3e}{6rr}+\frac{a^5e}{120r4}-\frac{a^7e}{5040r6}+$ &c. $=bq$.

Si iam in hac ſerie omnes termini, excepto primo, evaneſcere intelligantur, reſultabit $(e+r)a=bq$, hoc eſt analogia AE : DN :: AQ : AB, quæ analogia iam commodam præbet approximationem arcus AB.

Sin vero duo, tres, vel plures ex primis ſeriei terminis aſſumantur, orientur æquationes trium, quinque, ſeptem, &c. dimenſionum, quæ quidem uſui Aſtronomico non videntur idoneæ. Quare progredior ad modum arithmeticum ſolvendi hoc idem problema.

5. In ſecunda figura, exprimat arcus AM anomaliam

liam mediam, sitque CE eccentricitas, & iungatur ME ductaque ex centro C semicirculi linea CN parallela ipsi EM, ducatur EN: dico angulum AEN esse anomaliam coæquatam quam proxime: accuratius vero invenietur hæc anomalia; si iam arcus NM vocetur a & radius AC $= r$, fiatque $x = \frac{aa}{6rr} - \frac{a^4}{120r^4} + \frac{a^6}{5040r^6}$ —&c. & angulus NEB $=$ NC$^2 \times ax$: BE^2; Angulus AEB erit anomalia coæquata

Nam trilineum NEB fiet $=$ segmento circuli inter arcum NM & eius subtensam contento, quare cum triangula NEC & NMC super eadem basi NC & inter easdem parallelas NC, ME constituta æqualia sint, erit Sector AMC $=$ trilineo ABE; atqui Sector AMC: Semicirc. AMDA :: AM : AMD; ergo etiam ANBE: Semic. AMDA :: AM: AMD. Hoc est in data ratione. Quod erat &c.

Quod ad angulum BEN attinet, invenietur is ex eo, quod segmentum circuli inter arcum MN eiusque chordam MN æquari debeat sectori BEN. Iam segmentum illud æquat semissem facti ex differentia arcus MN eiusque sinus in radium CN. Verum cum arcus MN (*hyp*) sit a, eiusque sinus $= a - \frac{a^3}{6rr} + \frac{a^5}{120r^4} - \frac{a^7}{5040r^6}$ + &c. erit arcus MN—sinu eius $= \frac{a^3}{6rr} - \frac{a^5}{120r^4} + \frac{a^7}{5040r^6}$ &c. (hoc est propter $x = \frac{aa}{6rr} - \frac{a^4}{120r^4} + \frac{a^6}{5040r^6}$ &c.) $= ax$. Sector vero BEN æquat semissem rectanguli ex NE in Bb, demissa ex B in EN perpendiculari; Sector

enim ob parvitatem anguli BEN inſtar trianguli rectilinei ſpectari poteſt, & perpendiculum B*b* inſtar arculi centro E radio EB deſcripti. Quum autem ſegmentum NM æquare debeat ſectorem BEN, erit EN × B*b* = *ax* in CN; adeoque arculus B*b* = CN × *ax* : EN. Eſt vero ut EB ad arculum B*b*, ita radius CN ad angulum BEN = $CN^2 \times ax$: EB × EN, vel quia EN & EB iam inſtar æqualium tractari poſſunt, erit ideo angulus BEN = $CN^2 \times ax$: EB^2; ut ponebatur.

Exempl. Sit cum *Keillio* CE = 0. 01691, exiſtente AC = 1, quæritur anomalia coæquata AEB, ſi anomalia media AM = 30°. Quare angulus ACM = 30°, & in Δ^{lo}. MCE invenietur angulus MEA = NCA = 29°31'. 38" = arcui AN; quare NM = 28'. 22". In triangulo vero NCE, invenitur angulus NEA = 29°. 3'. 7", & hæc eſt anomalia coæquata; nam angulus NEB non æquabit 4 minuta quinta, adeoque attendi non meretur.

Tandem dicatur, ut ſinus totus ad ſinum anguli PCE (erecta nempe in E normali ad AD) ita tangens anguli inventi AEN vel AEB ad anomaliam coæquatam in Ellipſi. Cl. Keil per ſuam ſeriem invenit angulum AEN, qualem hoc loco absque ullo uſu ſeriei per ſolam reſolutionem binorum triangulorum MCE & NCE, aſſecuti ſumus, uſus enim præcedentis ſeriei mihi tantum monſtravit, quantus debeat eſſe angulus NEB addendus iam invento NEA? cum vero is ſubducto calculo ne quidem ad 4 minuta quinta accedere inveniatur, merito contemnitur.

DE

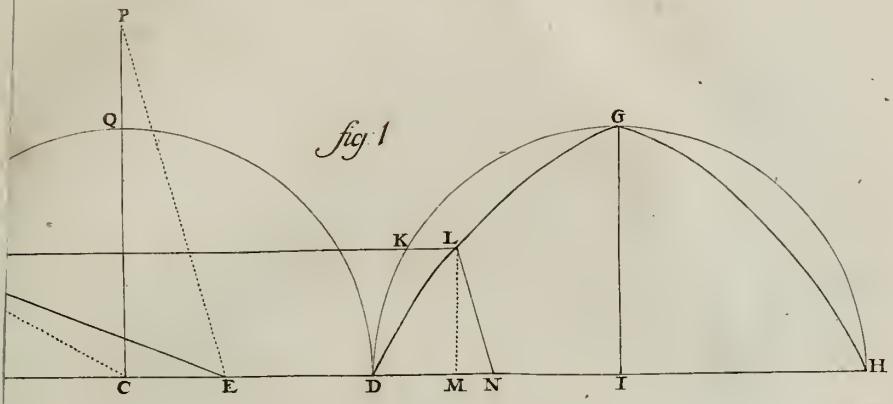
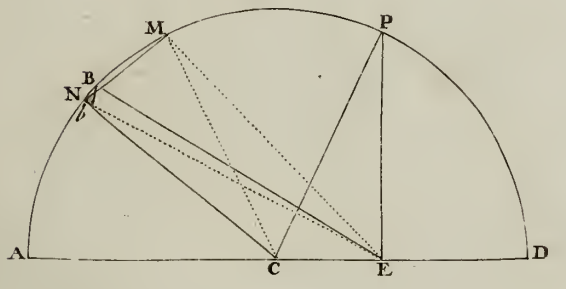


fig: 1

fig: 2.



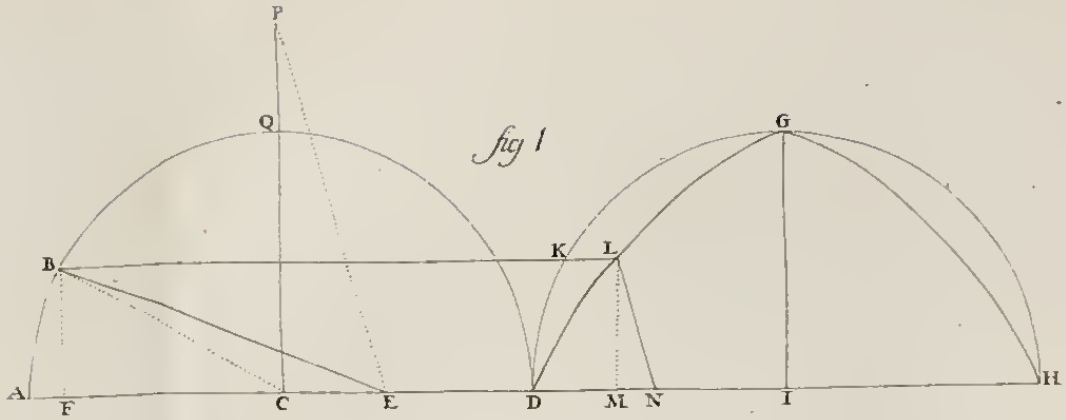


fig 1

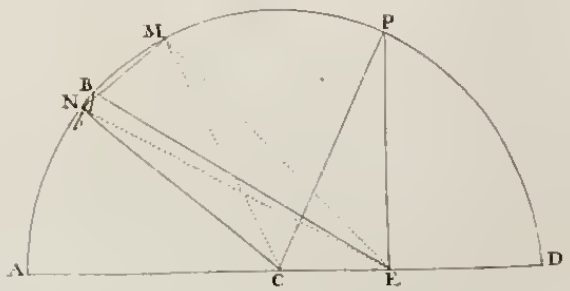


fig 2